Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-9166
Project Name	AI fit track API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	yes	Uniform formatting improve readability
2	Proper File Structure	Files and folders are logically organized	yes	Files are organized logically
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Clear names improve code understanding
4	Function / Method Names	Functions are descriptively named	yes	Descriptive names identify functionality clearly
5	Code Comments	Inline and block comments explain logic	yes	Important logic is explained With comments
6	Modular Design	Code is split into reusable functions/m dules	yes	Reusable modules make the code maintained
7	No Redundant Code	Duplicate or unused code is removed	yes	Code duplication is avoided
8	Error Handling	Exceptions and errors are handled gracefully	yes	Error are handled properly

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)

1	Authentication module	Node.js/Express.js/ JWT	Login,registration and protected API routes	high

2	Blog Management module	Node.js/Express.js	Create,read,update and delete blog posts	high
3	AI Content Generation module	Google Gemini API	AI-based blog generation and content assistance	high
4	API Error Handling module	Express.js	Used across API endpoint for consistence error response	high
5	Database module	MongoDB	User,blog and related data operation	high

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Code is logically organized ito separate modules and folders
Readability	5	Meaningful names and consistent formatting improve readability
Reusability	5	Common functionality is implemented using reusable module and function
Documentation / Comments	5	Modular structure makes the application easy to maintain and extend
Overall Score	4	Errors are handles using validation and consistent error responses